

Test Plan Documentation

Subgroup 3: Devices

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Revision History

Date	Version	Description	Author
08/May/26	1.0	Updated file to contain Table of content and figures.	Mustafa Al-Bayati, Sergej Macut, Tony Nguyen, Ryad Hazin
08/May/26	1.1	Initial draft of the test plan, defining unit testing and documentation scope for the ARDUINO_REFACTOR project.	Mustafa Al-Bayati, Sergej Macut, Tony Nguyen, Ryad Hazin

Test Plan

Task Delegation

Here is a balanced breakdown of the unit testing and documentation workload, divided by the folders and files shown in your `src` directory:

- **Ryad** will be responsible for the core logic, configuration, and system safety. This includes writing unit tests and documentation for the `core/` folder (`CommandHandler.cpp`, `CommandHandler.h`, `Observer.h`, `Subject.h`), the `config/` folder (`config.cpp`, `config.h`), and the `safety/` folder (`GasSafety.cpp`, `GasSafety.h`) Figure[3].
- **Mustafa** will handle the hardware device representations. Due to the size of the folder, Mustafa will focus entirely on the `devices/` folder, covering tests and documentation for `BuzzerDevice`, `Devices` (base/manager), `DoorDevice`, `Fan`, and `WindowDevice` figure[4].
- **Sergej** will take charge of the system inputs and safety mechanisms. This involves unit testing and documenting the `sensors/` folder (`sensors.cpp`, `sensors.h`, `SteamSensor.cpp`, `SteamSensor.h`), the `music/` folder (`MusicEngine.cpp`, `MusicEngine.h`), and the entry point file (`main.cpp`) Figure[2].
- **Tony** will manage the outputs, media, and the main execution loop. Tony will write tests and documentation for the `display/` folder (`LCDManager.cpp`, `LCDManager.h`), and the `gateway.js` file.

Figures

```
===== SUMMARY =====
Environment  Test      Status  Duration
-----
uno          test_music SKIPPED
uno          test_sensors SKIPPED
native       test_music PASSED    00:00:01.383
native       test_sensors PASSED    00:00:01.426
===== 8 test cases: 8 succeeded in 00:00:02.809 =====
```

Figure 2: Music and Sensors tests

```
===== SUMMARY =====
Environment  Test      Status  Duration
-----
uno          test_config PASSED    00:00:09.332
uno          test_core PASSED    00:00:10.646
uno          test_safety PASSED    00:00:09.166
===== 10 test cases: 10 succeeded in 00:00:29.144 =====
```

Figure 3: Config, Core and Safety tests

```
musta@Zeus MINGW64 ~/interactive-house/SG3_Devices/arduino_refactor (sg3-ut)
$ python -m platformio test -e native
Verbosity level can be increased via '-v, -vv, or -vvv' option
Collected 1 tests

Processing test_devices in native environment
Building...
Testing...
test\test_devices\test_main.cpp:18: test_buzzer [PASSED]
----- native:test_devices [PASSED] Took 1.06 seconds -----

===== SUMMARY =====
Environment  Test      Status  Duration
-----
native       test_devices PASSED    00:00:01.062
===== 1 test cases: 1 succeeded in 00:00:01.062 =====

musta@Zeus MINGW64 ~/interactive-house/SG3_Devices/arduino_refactor (sg3-ut)
```

Figure 4: Devices testing

```
esting -vvv
```

```
Flash: [= ] 11.7% (used 3774 bytes from 32256 bytes)
```

```
.pio\build\uno\firmware.elf :
```

section	size	addr
.data	370	8388864
.text	3404	0
.bss	223	8389234
.comment	17	0
.note.gnu.avr.deviceinfo	64	0
.debug_aranges	256	0
.debug_info	3020	0
.debug_abbrev	1602	0
.debug_line	1290	0
.debug_str	520	0
Total	10766	

```
----- uno:* [PASSED] Took 1.38 seconds -----
```

```
===== SUMMARY =====
```

Environment	Test	Status	Duration
uno	*	SKIPPED	00:00:01.385
native	*	SKIPPED	

```
===== 0 test cases: 0 succeeded in 00:00:01.385 =====
```